

SciOdyssey.

A research layer over your agent harness.

PRESERVE THE RESEARCH WORLD · RESET THE RESEARCHER

ROBUST

SUPERVISABLE

BROAD SEARCH

SAME EVIDENCE
FRESH JUDGMENT
A WIDER SEARCH

Research
Runs
Longer.

01 / PROBLEM

Why current agents fall short.

Sustained autonomous research breaks in three different ways.

01 Closed-world fragility

Unexpected runtime failures can halt an unattended workflow.

Cycle 1 · `numpy.float32` serialization error · autonomously repaired

02 Trajectory momentum

A single long-running trajectory keeps moving in the same direction.

Direct Codex control reached ~**0.6046** after extended local refinement

03 Context inflation cost

Reconstructing project context can cost far more than the edit itself.

Project-shaped replication · 340,087 vs 42,377 inclusive tokens · **8.03x**

DESIGN RESPONSE

Keep the world. Reset the trajectory.

02 / SYSTEM

One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

DUAL-LIFETIME LOOP

Persistent world · ephemeral trajectory · audited write-back

Fresh Scientist

Owns scientific judgment inside one trajectory: hypotheses, code, experiments, interpretation, and pivots.

META Review

Audits claims against artifacts, scopes conclusions, and decides what is durable enough to survive.

SELECTIVE PERSISTENCE

Only audited empirical findings, code state, and bounded priors write back to the shared world.

PERSISTENT RESEARCH WORLD

What survives across trajectories

A · EVIDENCE

Knowledge + Ledger

Metrics, failures, reports, and experiment receipts in `ledger.jsonl`.

B · PRIORS

Brief + Hypotheses

Current understanding and exploration leads in `brief.md`, open to revision.

C · STATE

Code + State

Retained implementation, serialized model weights, checks, and git provenance.

CONTEXT RESET

The next Scientist reads verified experience from disk, but does not inherit the previous reasoning trace.

RUNTIME SUBSTRATE

WORKTREE ISOLATION

Independent git worktrees isolate branches and preserve reproducible artifacts.

SELF-HEALING

Traceback → patch → rerun keeps unattended research moving after runtime faults.

PARALLEL SYNTHESIS

Completed branches can be compared post-hoc; useful ideas survive even when a branch loses.

The next Scientist inherits **verified experience** — not prior reasoning momentum.

Decoupled Memory · Audited Persistence · Fresh Scientific Judgment

03 / EVALUATION

Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

01 MEASURED RESULT

The model got better.

0.6016000

Official FM baseline

→

0.6059363

SciOdyssey submission

+0.0043363 Primary improvement

GAUC **0.6728421**

(+0.0054)

nDCG@5 **0.5390304**

(+0.0033)

LogLoss **0.38102**

(-0.0032)

Feature Engine: 46 categorical cross-interactions + log1p transforms

Architecture: 8-seed Factorization Machine ensemble (NumPy / CPU)

Monotonic gain from baseline (0.6016) → feature engineering (0.6041) → seed ensembling (0.6059); **100% CPU inference** without GPU clusters.

02 SUSTAINED SEARCH

The search stayed productive.

0.60163

C1: Valid Base

0.60409

C2: Features

0.60443

C3: Var. Red.

0.60594

C4: 8-Seed FM

SciOdyssey	0.6059363	Climbed across 4 cycles (7 Full evals)
Codex Goal	0.6044533	Peaked at C8; stalled & dropped in 22 runs

4 autonomous cycles

7 Full evaluations

+0.00148 over Codex

7 Full public-validation evaluations retained across 4 cycles. Trajectory climbed via **structural feature expansion** rather than single lucky edits.

03 SELF-HEALING & GATEKEEPING

The agent recovered from failure.

TIER 1 · RUNTIME SELF-HEALING	CRASH AUTO-RECOVERY
FAILURE ACTION	Cycle 1 <code>numpy.float32</code> serialization crash Scientist parsed traceback, auto-patched scalar helper
OUTCOME	0 human bugfixes · Unattended rerun succeeded

TIER 2 · META GATEKEEPING	SPURIOUS STATE REJECTION
FALSE CLAIM ACTION	Scientist claimed BPR gain on medium proxy META audited system/data.py, caught UNK leak
OUTCOME	Promotion blocked · Zero contaminated state

Empirical receipts preserved: runtime self-healing telemetry and independent code-level proxy audits.

04 ARCHITECTURE ABLATION

Meta-scientist beats direct agents.

01 · DIRECT Antigravity	02 · SUBAGENTS + Subagents	03 · META-SCIENTIST Persistent World
0.60458	→ 0.60472	→ 0.60520
Single 2h run	Delegated swarm	+ fresh Scientist

+0.00062 organization gain (Gemini 3.7 Flash)

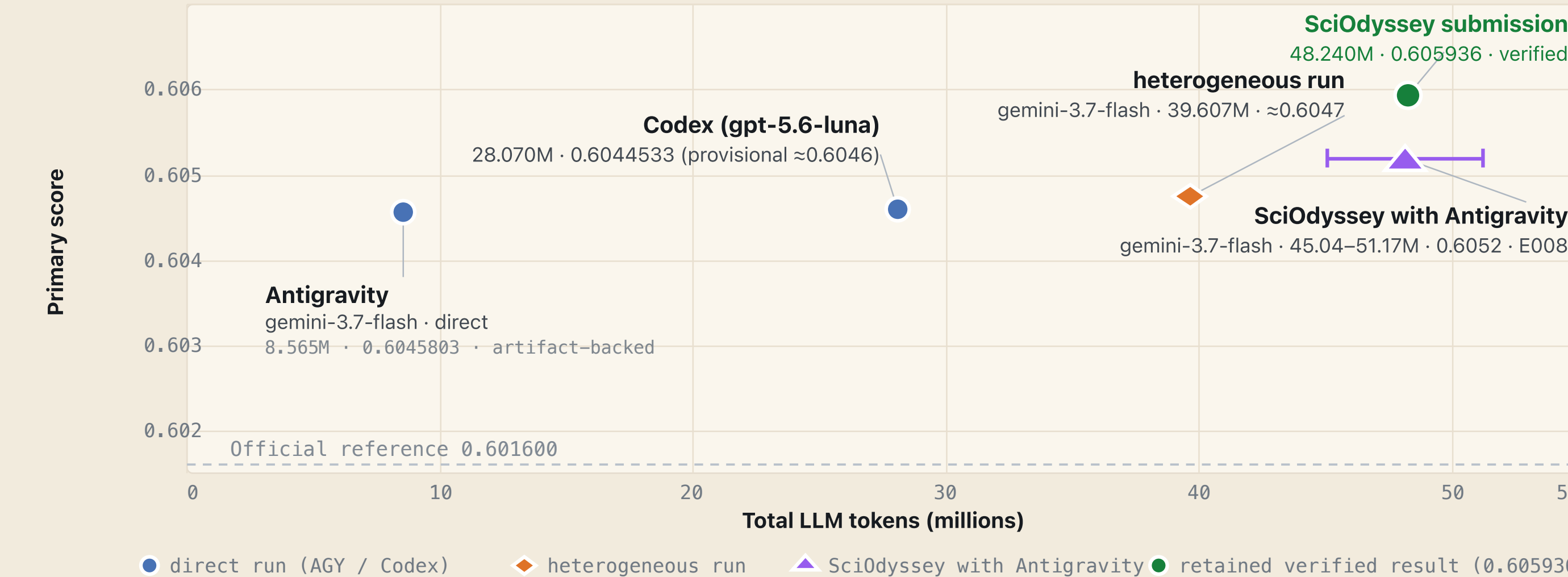
Keep Gemini 3.7 Flash fixed and change only research organization: **direct agent (0.6046) → subagents (0.6047) → Meta-Scientist (0.6052)**.

05 RESOURCE ACCOUNTING

We are not tokenmaxxing. Score vs token investment.

~\$10 cost · 1 × M2 · 0 GPU-hr · 1h 55m

KuaiRand-Pure · Public validation · Inclusive input+output tokens



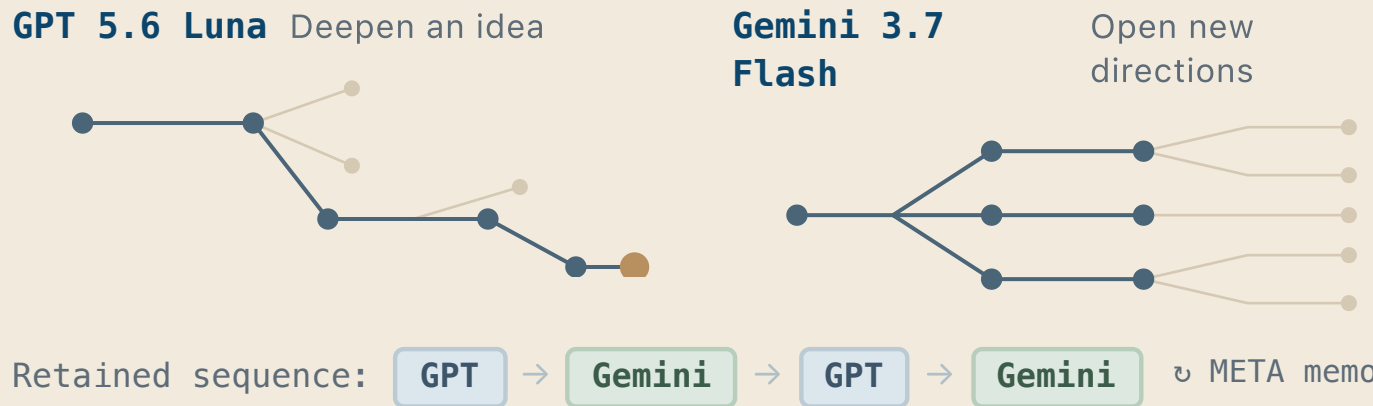
All results **verified** on KuaiRand-Pure official validation split with **complete reproducible artifacts**.

04 / INSIGHTS

What the run taught us.

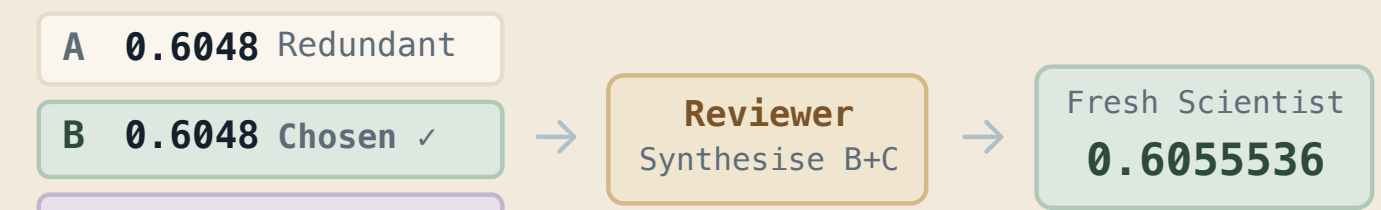
01 · MODEL DIVERSITY

Depth + Breadth Synergy.



02 · PARALLEL SYNTHESIS

Cross-Branch Idea Transfer.



Lower-scoring Branch C contributed pairwise ranking and rank normalisation — a lower score can still carry a useful idea.

Branch B+C Ideas → Synthesis **0.6055536**

05 / EXPERIENCE

Start your journey.

task.md

What to solve

Formal task contract: research objectives, dataset inputs, compute constraints, and official evaluation protocol.

- Dataset paths & split integrity
- Official sandbox evaluator

PERSONAL.md

How to work

User research preferences, model guidance, exploration style, tool permissions, and supervisory checkpoints.

- Model rotation (GPT / Gemini)
- Supervision & budget thresholds

SUPERVISION TIERS

Tier 1 Unattended Overnight Runs · **Tier 2** Checkpoint Review (META) · **Tier 3** Steer Working Priors

1. Define task.md → 2. Single-step test → 3. Autonomous search → 4. Parallel explore → 5. Inspect evidence

DECLARATIVE PROTOCOL

Human defines boundary and preferences (task.md + PERSONAL.md); SciOdyssey autonomously iterates, heals, and crystallizes verified evidence.

Autonomous by default.

Supervisable by design.

github.com/zc6600/research-agent-kuairand

Team Good4AI: Chen Zhu · Zhou Ziyu · Shilin Xu · GE GAO · Jiran Li

Sources: slides/slides.md · docs/ARCHITECTURE.md · docs/FINAL_REPORT.md · KuaiRand-Pure submission ledger

TEAM GOOD4AI · ROBUST · SUPERVISABLE · BROAD SEARCH

A Persistent Research World for Autonomous Science

PROJECT WIKI & DEMO

sciodyyssey.zc6600.wiki

Scan for live report & artifacts

